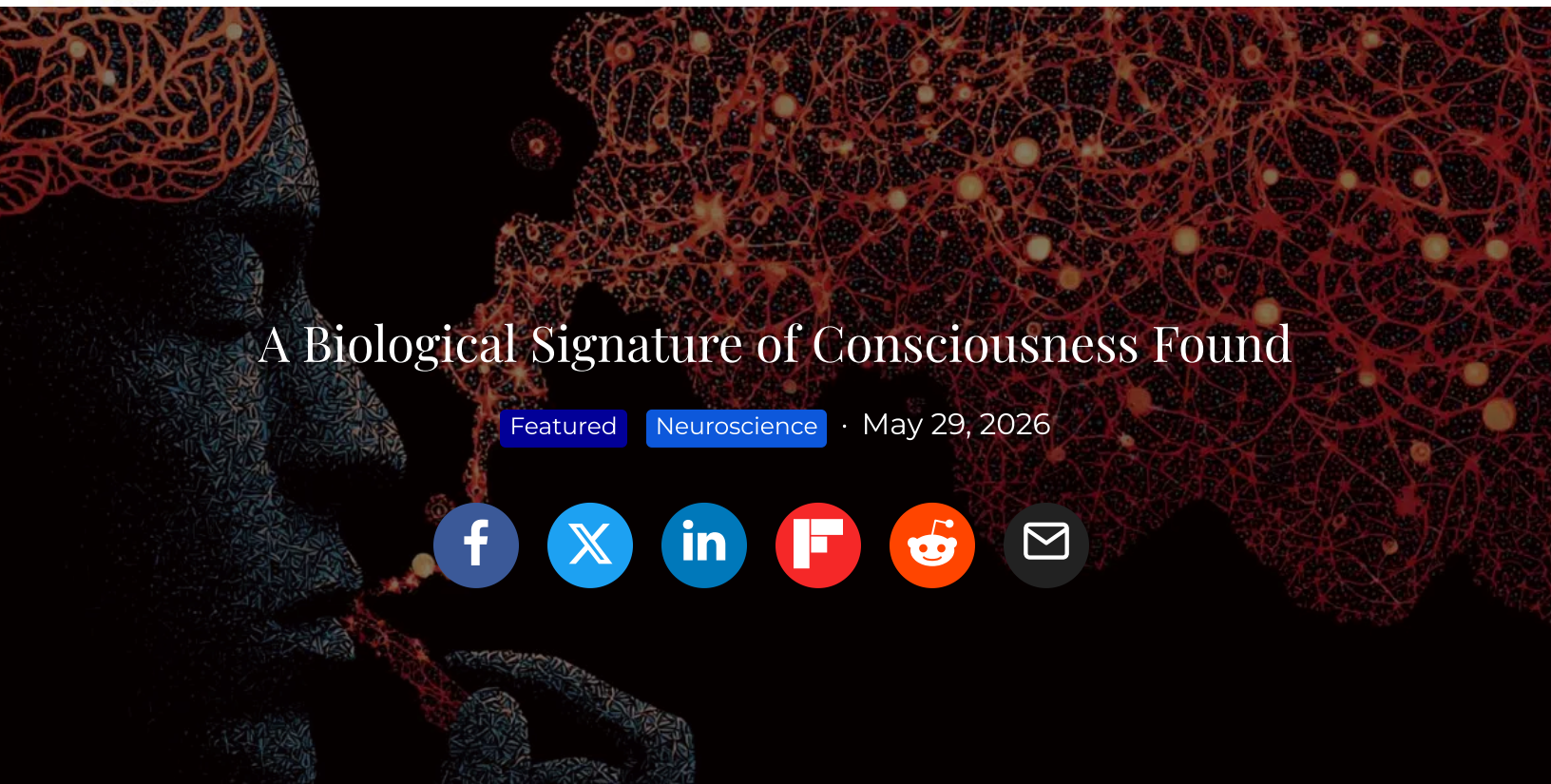




A Biological Signature of Consciousness Found

Featured Neuroscience · May 29, 2026



A rapid 20–45 Hz oscillation within the central human thalamus serves as a distinct biological signature of conscious awareness, occurring exclusively during waking hours and REM sleep while falling completely silent during non-REM cycles. Credit: Neuroscience News

Summary: A precision neuropsychology and clinical neurology study unmasked a hidden bio-physiological rhythm deep within the human midbrain that encodes active states of consciousness. The research documents a previously unknown rapid activity pattern operating in the human thalamus, the brain's central gateway for perception and attention.

By capturing direct, intracranial electrical recordings from patients undergoing deep brain stimulation, the investigative team isolated a

highly specific 20 to 45 Hertz (Hz) oscillation that triggers exclusively during conscious waking hours and vivid REM sleep, vanishing completely during unconscious non-REM sleep cycles.

Key Facts

- **The Sentinel of Perception:** The thalamus is a deep-lying structure situated at the absolute center of the brain, functioning as a master relay station that gathers and routes signals across diverse cortical networks. It serves as an internal gate for human perception and attention, playing a foundational role in sustaining conscious awareness.
- **The Waking 20–45 Hz Identifier:** Led by Professor Tobias Staudigl and PD Dr. Elisabeth Kaufmann, the LMU team discovered a rapid thalamic oscillation pulsing precisely in the 20 to 45 Hertz frequency range. This micro-rhythm acts as an absolute biological signature of conscious awareness: it occurs exclusively when a subject is fully awake or immersed in high-intensity REM dream states.
- **The Non-REM Blackout:** The newly isolated rapid signature disappears during non-REM sleep phases, when eye movements cease and conscious experience is heavily diminished. During these periods of reduced consciousness, the 20–45 Hz thalamic rhythm falls completely silent, overtaken instead by slow-wave delta brain oscillations.

- **Implanted Electrode Direct Capture:** Because recording deep brain structures is notoriously difficult using standard surface EEGs, researchers partnered with epilepsy patients undergoing deep brain stimulation therapy. By utilizing surgical electrodes implanted directly into the central thalamus, scientists gained a rare, pristine window to record direct local field potentials.
- **Multi-Modal Data Integration:** To ensure absolute diagnostic accuracy, lead author Dr. Aditya Chowdhury combined the direct deep-thalamic recordings with external surface EEGs, continuous eye-movement tracking, and micro-classified patient sleep logs. This allowed the team to track exactly how midbrain frequencies fluctuate second-by-second across varying states of awareness.
- **A Clinical Gateway to Neurological Repair:** Characterizing this signature opens up clinical avenues to optimize existing deep brain therapies. Backed by newly awarded funding from the European Research Council, the LMU laboratory is moving forward to leverage this thalamic rhythm to map, monitor, and treat an array of complex neurological disorders.

Source: LMU

Neuropsychology researchers at LMU have discovered a rhythm in the midbrain that could serve as a bio-physiological signature for specific states of consciousness.

The thalamus is a deep-lying structure in the center of the brain which gathers and relays signals from many different areas of the brain. It functions like a gate for perception and attention and is thought to play a key role in supporting conscious states.

In a study published recently in the journal *Nature Human Behaviour*, Professor Tobias Staudigl (Psychology, LMU) and his team, in collaboration with PD Dr. Elisabeth Kaufmann (Neurology, LMU), have discovered a previously unknown rapid activity pattern in the human thalamus.

This rapid oscillation, in the frequency range of 20 to 45 Hertz, occurs exclusively during waking hours and REM sleep, the phase of sleep with rapid eye movements and intensive dreams. It is entirely absent in non-REM sleep, when eye movements are absent and consciousness is strongly reduced. In this sleep phase, the brain activity is dominated instead by slower oscillations.

Measurements with implanted electrodes

In this study, the researchers investigated patients undergoing deep brain stimulation therapy, a form of treatment for epilepsy. The therapy involves implanting electrodes in the thalamus of the patients to reduce the number of epileptic seizures.

From a scientific perspective, this offers an exciting and very rare opportunity to directly record neural activity in the human thalamus. Recording neural activity from such deep brain structures is notoriously difficult using common methods such as surface EEG.

The researcher's findings were based on direct field potential recordings in the central thalamus, combined with surface EEG measurements, eye movement analyses, and the classification of sleep patterns in individual patients. This allowed them to precisely track how thalamic oscillation patterns change when the subjects were awake or in various sleep phases.

“Our results show that the central thalamus plays an important role in regulating brain states. In the context of existing research, our results show that this small deep-lying brain structure could actively influence our states of consciousness,” explains Dr. Aditya Chowdhury, lead author of the study.

Tobias Staudigl adds: “These characteristic rhythm patterns can be reliably attributed to specific states and thus have the potential to serve as a measurable biological signature of states of consciousness.”

A deeper understanding of the signal discovered in the thalamus would also be of much interest from a clinical perspective. The signal could be used to optimize existing therapeutic approaches and, in the

long run, lead to new approaches for the treatment of other neurological diseases.

Funding: Staudigl was recently awarded funding by the European Research Council to further explore the clinical potential of the discovery.

Key Questions Answered:

Q: Why did scientists have to look deep inside the center of the brain instead of just scanning the scalp to find this consciousness switch?

A: Because the master switchboard of awareness, the thalamus, is buried too deep beneath layers of bone, muscle, and outer brain tissue for normal surface equipment to read. Standard scalp EEGs easily capture broad, surface-level signals, but they completely miss the faint, high-frequency rhythms of the midbrain. By working with epilepsy patients who already required surgically implanted deep-brain electrodes, LMU researchers bypassed the noise to listen directly to the central gateway of perception.

Q: Why does the exact same 20–45 Hz rhythm light up when we are wide awake *and* when we are dead asleep dreaming?

A: Because dreaming during REM sleep is actually an intense, hyper-active state of consciousness. Even though your body is paralyzed in bed, your mind is actively generating vivid experiences, processing

deep emotions, and scanning visual environments in your mind's eye. The thalamus treats active waking life and active dreaming life as nearly identical processing tasks, engaging the exact same 20–45 Hz fast-pulsing gate for both.

Q: How can isolating a tiny midbrain rhythm help doctors treat complex neurological illnesses in the future?

A: It provides an objective, real-time biological map of a patient's exact state of consciousness. Instead of guessing how deeply a neurological disease has impaired a patient's cognitive networks, doctors can monitor this 20–45 Hz signature. In the long run, clinicians can use this signal to program smart, responsive deep-brain stimulation implants that automatically dial their electrical pulses up or down to repair damaged consciousness loops.

Editorial Notes:

- This article was edited by a Neuroscience News editor.
- Journal paper reviewed in full.
- Additional context added by our staff.

About this consciousness and neuroscience research news

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[“Thalamic oscillations distinguish natural states of consciousness in humans”](#) by Aditya Chowdhury, Xiongbo Wu, Tara Beilner, Thomas Schreiner, Thomas Koeglsperger, Jan-Hinnerk Mehrkens, Jan Remi, Christian Vollmar, Elisabeth Kaufmann & Tobias Staudigl. *Nature Human Behaviour*

DOI:10.1038/s41562-026-02446-z

Abstract

Thalamic oscillations distinguish natural states of consciousness in humans

Natural states of consciousness are thought to be regulated by deep brain structures such as the thalamus. However, very little is known about the underlying electrophysiology in humans.

Here, using a rare opportunity to directly record from the human thalamus, we identify a hitherto-unreported brain-state-specific oscillation of approximately 19–45 Hz. This oscillation is present

only during rapid eye movement (REM) sleep and wakefulness, while being absent during non-REM sleep.

The 19–45 Hz oscillation further distinguishes REM sleep microstates, co-occurring with bursts of eye movements, and is specific to the central thalamus, a structure implicated in causing global brain state transitions.

The discovery of a distinct oscillatory signature in the central thalamus that distinguishes conscious states opens up avenues to further investigate thalamic contributions to states of consciousness in humans and potentially to refine interventions to treat disorders of consciousness.

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Dr Simant Kumar Nanda

[Reply](#)

June 1, 2026 at 8:31 am

It is understandable that dopamine is the culprit in stress induced sexual dysfunction

Amrendra Kumar

Reply

May 30, 2026 at 10:26 am

“Biosignature of Consciousness” must be replaced by “Biosignature for Consciousness” because Consciousness is a process for converting alertness into Awareness.

Grant Castellou

Reply

May 30, 2026 at 10:01 am

It's becoming clear that with all the brain and consciousness theories out there, the proof will be in the pudding. By this I mean, can any particular theory be used to create a human adult level conscious machine. My bet is on the late Gerald Edelman's Extended Theory of Neuronal Group Selection. The lead group in robotics based on this theory is the Neurorobotics Lab at UC at Irvine. Dr. Edelman distinguished between primary consciousness, which came first in evolution, and that humans share with other conscious animals, and higher order consciousness, which came to only humans with the acquisition of language. A machine with only primary consciousness will probably have to come first.

What I find special about the TNGS is the Darwin series of automata created at the Neurosciences Institute by Dr. Edelman and his colleagues in the 1990's and 2000's. These machines perform in the real world, not in a restricted simulated world, and display convincing physical behavior indicative of higher psychological functions necessary for consciousness, such as perceptual categorization, memory, and learning. They are based on realistic models of the parts of the biological brain that the theory claims subserve these functions. The extended TNGS allows for the emergence of consciousness based only on further evolutionary development of the brain areas responsible for these functions, in a parsimonious way. No other research I've encountered is anywhere near as convincing.

I post because on almost every video and article about the brain and consciousness that I encounter, the attitude seems to be that we still know next to nothing about how the brain and consciousness work; that there's lots of data but no unifying theory. I believe the extended TNGS is that theory. My motivation is to keep that theory in front of the public. And obviously, I consider it the route to a truly conscious machine, primary and higher-order.

My advice to people who want to create a conscious machine is to seriously ground themselves in the extended TNGS and the Darwin automata first, and proceed from there, by applying to Jeff

Krichmar's lab at UC Irvine, possibly. Dr. Edelman's roadmap to a conscious machine is at <https://arxiv.org/abs/2105.10461>, and here is a video of Jeff Krichmar talking about some of the Darwin automata, <https://www.youtube.com/watch?v=J7Uh9phc1Ow>

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